

Design and Spectral Engineering of an 8-Channel WDM Demultiplexer Based on Cascaded Mach-Zehnder Interferometers

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ABSTRACT Wavelength Division Multiplexing (WDM) is one of the most important technologies in modern optical communication systems due to its ability to transmit multiple optical channels through a single waveguide. In silicon photonics, cascaded Mach-Zehnder interferometers (MZIs) provide a compact and controllable solution for implementing wavelength multiplexing and demultiplexing systems. This work presents the design and analysis of an 8-channel WDM demultiplexer based on cascaded MZI stages. The spectral evolution across the network is studied by progressively engineering the free spectral range (FSR) and phase response of each stage. A matrix-based transfer function formalism is employed to derive the spectral response of the complete architecture. Additionally, thermal phase shifters are introduced to selectively position transmission windows and compensate fabrication deviations. Simulation results demonstrate how cascaded interferometric filtering enables accurate wavelength channel separation.

INDEX TERMS Silicon photonics, wavelength division multiplexing, Mach-Zehnder interferometer, integrated optics, photonic integrated circuits, spectral filtering, thermo-optic phase shifters.

I. INTRODUCTION

THE continuous increase in bandwidth demand has driven the development of high-capacity optical communication systems. Wavelength Division Multiplexing (WDM) enables multiple optical carriers operating at different wavelengths to propagate simultaneously through a single optical waveguide, significantly increasing transmission capacity without increasing the physical infrastructure.

Silicon photonics has emerged as a promising platform for WDM systems due to its compatibility with CMOS fabrication processes, reduced power consumption, and high integration density. Several architectures have been proposed for implementing integrated WDM systems, including Arrayed Waveguide Gratings (AWGs), micro-ring resonators, Bragg gratings, and Mach-Zehnder interferometer (MZI) networks [1].

Among these alternatives, cascaded MZI architectures provide an attractive trade-off between spectral control, insertion loss, scalability, and fabrication simplicity. Furthermore, the inclusion of thermo-optic phase shifters enables post-fabrication tuning and compensation for process variations [2]. This paper presents the theoretical formulation, architectural design, and spectral analysis of an 8-channel WDM demultiplexer implemented using cascaded MZIs.

II. Case Study Context

The analyzed system corresponds to a integrated photonic WDM demultiplexer composed of three cascaded MZI stages. The structure progressively refines the spectral response to separate eight wavelength channels, figure 1.

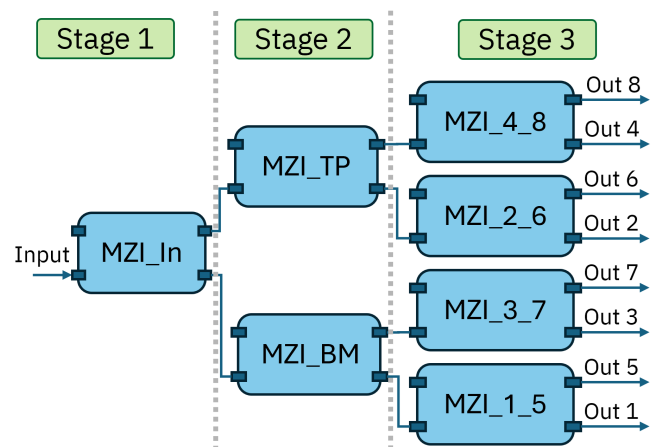


FIGURE 1. Cascaded MZI network used for 8-channel WDM demultiplexing. The architecture is organized into three interferometric stages that progressively split and filter the optical spectrum.

The architecture follows a hierarchical filtering approach:

- Stage 1 (MZI_{In}) performs the initial spectral decomposition of the input optical signal into two complementary wavelength groups corresponding to odd and even output channels. This first interferometric division establishes the global spectral partitioning of the WDM network.
- Stage 2 further subdivides these spectral groups into complementary channel pairs to simplify the final wavelength extraction process. The odd-channel branch is separated into the (1, 5) and (3, 7) channel groups through MZI_{BM} , while the even-channel branch is divided into the (2, 6) and (4, 8) groups through MZI_{TP} . This intermediate decomposition reduces spectral overlap and increases filtering selectivity prior to the final stage.
- Stage 3 employs four independent output MZIs (MZI_{1_5} , MZI_{2_6} , MZI_{3_7} , and MZI_{4_8}) to isolate the individual wavelength channels. By exploiting the complementary bar and cross responses of each interferometer, the architecture enables selective routing of each wavelength toward its corresponding output port.

The topology enables the generation of eight distinct transmission windows over the wavelength range between 1500 nm and 1600 nm.

III. PROPOSED APPROACH

The MZI structure is modeled using matrix formalism. Each interferometer consists of:

- Two multimode interferometers (MMIs)
- One differential delay section
- One phase shifter

The transfer matrix of an ideal 50:50 MMI coupler is:

$$U = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & j \\ j & 1 \end{bmatrix} \quad (1)$$

The phase shifter matrix is given by:

$$P = \begin{bmatrix} e^{j\phi} & 0 \\ 0 & 1 \end{bmatrix} \quad (2)$$

The propagation delay matrix becomes:

$$D = \begin{bmatrix} e^{-j\beta L} & 0 \\ 0 & 1 \end{bmatrix} \quad (3)$$

where:

- ϕ is the applied phase shift
- β is the propagation constant
- L is the differential optical path length

The overall MZI transfer matrix is therefore:

$$H = U \cdot P \cdot D \cdot U \quad (4)$$

which results in:

$$H = \frac{1}{2} \begin{bmatrix} PD - 1 & j(1 + PD) \\ j(1 + PD) & 1 - PD \end{bmatrix}$$

where:

$$P = e^{j\phi} \quad (5)$$

$$D = e^{-j\beta L} \quad (6)$$

Considering the lower port of the input MZI as the excitation input of the WDM network, and following the mathematical development presented in this section, the resulting transfer functions for output channels 1 and 5 are obtained as:

$$E_{out_5}^{Cross} = \frac{j}{8}(1 - D)(1 - P_{BM}D_1)(P_{1_5}D_2 + 1) \quad (7)$$

$$E_{out_1}^{Bar} = \frac{1}{8}(1 - D)(1 - P_{BM}D_1)(1 - P_{1_5}D_2) \quad (8)$$

The responses associated with output channels 3 and 7 are:

$$E_{out_7}^{Cross} = \frac{j^2}{8}(1 - D)(P_{BM}D_1 + 1)(P_{3_7}D_2 + 1) \quad (9)$$

$$E_{out_3}^{Bar} = \frac{j}{8}(1 - D)(P_{BM}D_1 + 1)(1 - P_{3_7}D_2) \quad (10)$$

The responses associated with output channels 2 and 6 are:

$$E_{out_6}^{Cross} = \frac{j^2}{8}(D + 1)(1 - P_{TP}D_1)(P_{2_6}D_2 + 1) \quad (11)$$

$$E_{out_2}^{Bar} = \frac{j}{8}(D + 1)(1 - P_{TP}D_1)(1 - P_{2_6}D_2) \quad (12)$$

And finally the responses associated with the output channels 4 and 8 are:

$$E_{out_8}^{Cross} = \frac{j^3}{8}(D + 1)(P_{TP}D_1 + 1)(P_{4_8}D_2 + 1) \quad (13)$$

$$E_{out_4}^{Bar} = \frac{j^2}{8}(D + 1)(P_{TP}D_1 + 1)(1 - P_{4_8}D_2) \quad (14)$$

A. Free Spectral Range Engineering

The Free Spectral Range (FSR) is defined as:

$$FSR = \frac{\lambda^2}{n_g \Delta L} \quad (15)$$

where:

- λ is the operating wavelength
- n_g is the group index
- ΔL is the path length difference

The cascaded stages employ different delay lengths:

The base delay length is calculated as:

TABLE 1. MZI Stage Configuration

Stage	ΔL	FSR
Stage 1	L	25 nm
Stage 2	$L/2$	50 nm
Stage 3	$L/4$	100 nm

$$L = \frac{\lambda^2}{n_g \cdot FSR} \quad (16)$$

Using an operating wavelength of $\lambda = 1550$ nm and the target free spectral range (FSR), the required path length difference L is approximately $23 \mu m$.

IV. Simulation Setup

The WDM network was simulated by cascading the transfer functions of all MZI stages. Each stage progressively filters the spectrum and reduces the number of transmitted channels.

The spectral responses were evaluated across the wavelength range:

$$1500 \text{ nm} \leq \lambda \leq 1600 \text{ nm}$$

Thermo-optic phase shifters were incorporated to align the transmission windows at the desired wavelengths.

The selected phase values are summarized in Table II.

TABLE 2. Phase Shift Configuration

Phase Shifter	Phase (rad)	Phase ($^\circ$)	FSR (nm)	$\Delta\lambda$ (nm)
PS_BM	$\pi/2$	90	50	12.5
PS_TP	0	0	50	0
PS_1_5	1.75π	315	100	87.5
PS_3_7	1.25π	225	100	62.5
PS_2_6	1.5π	270	100	75
PS_4_8	π	180	100	50

V. Experimental Results and Analysis

A. Complementary Port Responses

One important characteristic of MZI-based filters is the complementary spectral behavior between the bar and cross output ports.

For a given interferometer, the transmission responses satisfy

$$|H_{Bar}|^2 + |H_{Cross}|^2 = 1 \quad (17)$$

which guarantees that the spectral window rejected at one port is transferred to its complementary output port.

This property plays a fundamental role in the proposed WDM architecture. Since the wavelength channels are hierarchically grouped across the cascaded MZI stages, selecting a transmission window at the bar output automatically ensures the complementary spectral response at the cross output.

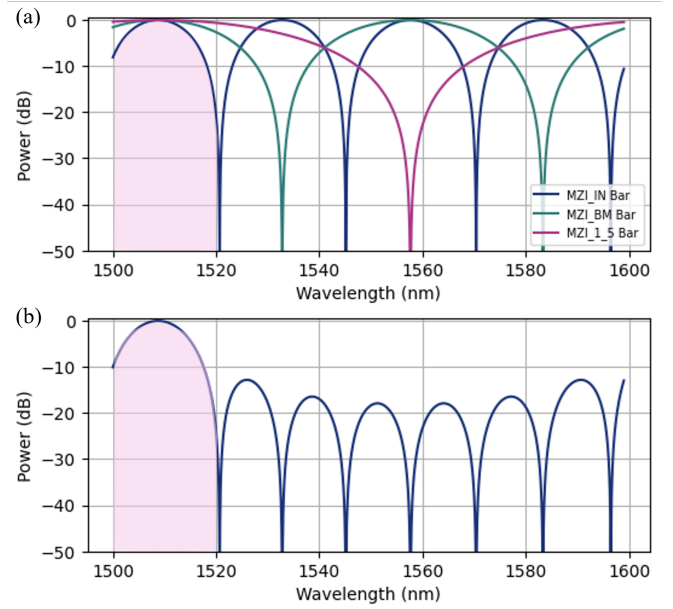


FIGURE 2. Construction of the first spectral transmission window of the cascaded WDM architecture. (a) Spectral alignment of the bar responses across the cascaded MZI stages, show how the transmission windows progressively overlap to refine the desired spectral region. (b) Final isolated transmission response corresponding to channel 1 after the cascaded filtering process.

For example, after properly tuning the thermo-optic phase shifters, if the transmission window corresponding to channel 1 is aligned at the bar output of MZI_{1_5} , the complementary response naturally routes channel 5 toward the cross output. This complementary filtering mechanism enables efficient wavelength separation while reducing the complexity of the spectral engineering process.

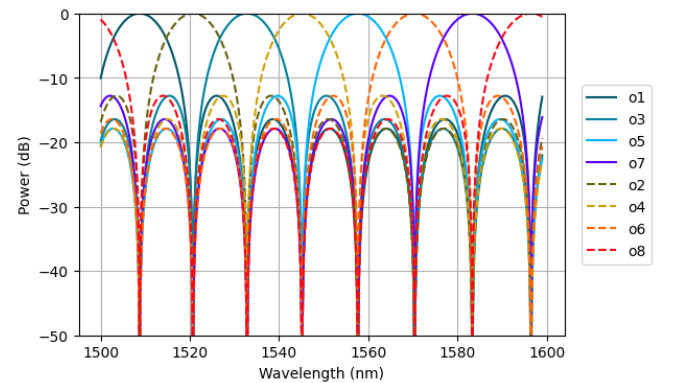


FIGURE 3. Spectral response of the 8-channel cascaded MZI-based WDM demultiplexer across the wavelength range from 1500 nm to 1600 nm.

B. Phase Shifting Effects

Thermal phase shifters enable precise positioning of the spectral transmission windows which allows dynamic tuning and compensation for fabrication deviations figure 2.

Simulation results demonstrate that appropriate phase selection enables the alignment of all eight transmission channels across the operating spectrum figure 3.

C. GDS Layout Implementation

Figure 4 shows the physical layout implementation for future fabrication of the proposed cascaded MZI-based WDM demultiplexer.

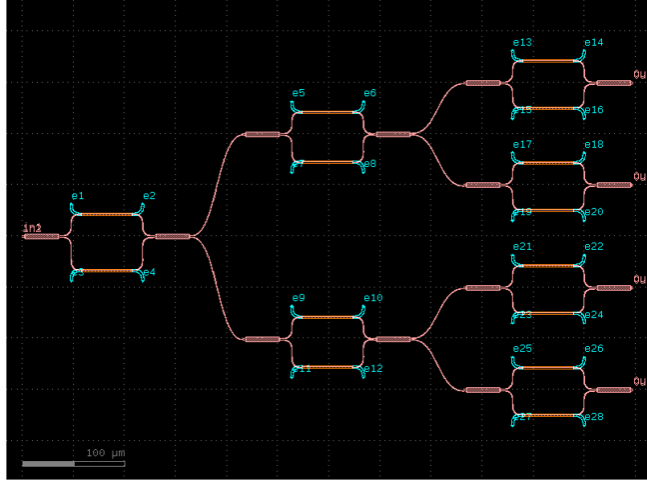


FIGURE 4. GDS layout implementation of the proposed 8-channel cascaded MZI-based WDM demultiplexer.

VI. Conclusion

An 8-channel WDM demultiplexer based on cascaded Mach-Zehnder interferometers has been presented and analyzed. The proposed architecture demonstrates how hierarchical spectral filtering can be achieved through successive interferometric stages with engineered free spectral ranges.

The matrix formalism provides a compact analytical framework for deriving the transfer functions of the complete network. Additionally, thermo-optic phase shifters enable spectral control and post-fabrication correction.

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